

Invited Opinion

The prevalence of sleep loss and sleep disorders in young and old adults

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ABSTRACT

The ability to sleep declines with age. The National Sleep Foundation, USA has recommended a minimum sleep amount for all ages. Individuals who experience sleep lesser than the recommended amount could be sleep-deprived. Several factors like stress, altered circadian cycle, medical conditions, etc. cause sleep deficiency. Almost 50–60 % of elderly population suffer from sleep disorders such as sleep apnea, restless legs syndrome, REM sleep behavior disorder, etc. Chronic sleep deprivation may further lead to the development of diseases such as Alzheimer's and Parkinson's. This paper reviews the prevalence of sleep disorders and consequences of sleep loss in young and old adults.

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Introduction

Sleep is a natural requirement of our body, during which the perception of sensory stimuli, consciousness, and voluntary muscle activity remains suppressed [31]. Electrophysiologically, two stages of sleep, Non-Rapid Eye Movement (NREM) sleep and Rapid Eye Movement (REM) sleep, can be distinctly characterized in a majority of mammals, including humans [10,71,72]. The functions of sleep are still not clearly known, but several theories have been put forward regarding them. Studies suggest that sleep is involved in various tasks at different levels. For example, sleep plays a role in the modulation of gene expression at the molecular level, synaptic transmission and synaptic architectural re-organization at the cellular level, metabolic and immunological processes at the physiological level, and behavioral and cognitive functions at the organizational level [66,35,62,88,89,153,36,90]. These studies suggest that sleep may be playing a multi-dimensional role

from cellular to organ levels for the homeostatic maintenance of our brain and body.

Although intermittent sleep disruption for a few hours to days could be experienced as nothing but a nuisance, however, chronic sleep deficit may induce severe complications in young as well as in old adults [70]. It may cause moodiness, fatigue, irritability, depression, forgetfulness, cognitive deficiency, increased appetite, carbohydrate cravings, reduced sex drive, etc. [138,139,155,132,39,68,3,70]. It could also be associated with frequent illness, obesity, diabetes, and hypertension, which alone or together lead to poor quality of life [159,70]. Therefore, it is necessary to understand how much time daily-one must spend in sleep. The 'National Sleep Foundation' (NSF), USA, has recommended a minimum daily sleep time for newborns, children, and young and old adults [64]. The NSF has recommended 14–17 h of sleep each day for 0–3 months old newborns and 12–15 h of sleep for 4–11 months old infants. The 3–5 years old preschoolers should sleep 10–13 h, whereas 6–13 years old school-goers should sleep 9–11 h a day. Similarly, a minimum daily sleep time of 7–9 h has been recommended for 18–25 years young adults and adults

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26–64 years old. For older adults (65 + years), the recommended minimum daily sleep time is 7–8 h [64].

The minimum daily sleep time in a night recommended by the NSF, USA, is for normal healthy individuals. Some subjects, however, tend to go to sleep and wake up early (from 8 p.m. to 4 a.m.) and are commonly known as “morning larks”. On the other hand “night owl” personality type may like to stay up late and wake up late (from 4 a.m. until 12 p.m.). The advanced or delayed sleep phases are merely descriptions of an individual's sleep schedule and do not represent pathological conditions. However, some genetic mutations alter the chrono-properties of the subjects and make them either familial natural short-sleepers or long-sleepers [105,131,11]. For example, the mutation in the transcriptional repressor gene of hypocretin ‘DEC2’ and the beta-1 adrenergic receptor gene ‘ADRB1’ leads to reduced sleep duration in the affected individuals. These individuals sleep only for 4–6 h per night and do not demonstrate daytime sleepiness or any signs of sleep deficits [105,131]. On the other hand, some people probably require more sleep than others; no genetic variations, however, have been detected in the familial natural long-sleepers [105,131,11]. People who are familial natural short-sleepers tend to have a very high behavioral drive and a need to be always busy with some suitable work. Additionally, it appears that these people have high pain thresholds and are quite resilient to many stressors, which can be attributed to a mutation in a specific gene [11].

Reduced or altered sleep patterns may be caused due to either physiological alteration and/or intentional or unintentional unusual behavioral manifestation [30,63]. For example, late-night TV watching or purposely going to bed late and wakening up early can contribute to sleep deficit. A sleep study was conducted from 2003 to 2011 using the Defense Meteorological Satellite Program's/Operational Linescan System (DMSP/OLS) for the measurement of outdoor evening light. It was observed that subjects living in the high outdoor evening light (as is found in big cities) demonstrated delayed bed- and wake-up time with reduced sleep episode duration and excessive daytime sleepiness [102]. Further these subjects showed poor sleep quality and sleep profiles very similar to a disorder associated with circadian rhythm [102]. It has been reported that both plasma melatonin concentrations and the entrained phase of the human circadian pacemaker were considerably influenced by the marginal variations in the exposed light for many days during the late evening hours [162]. Hence, it is likely that altered plasma melatonin levels would affect sleep timing as well as sleep quality [27,70]. Many electronic devices such as televisions, smartphones, tablets, gaming devices, fluorescent light bulbs, LED bulbs, computer monitors, etc. emit blue rays (400 to 490 nm wavelengths) and studies suggest that blue LED irradiance significantly suppresses melatonin secretion and alters sleep [157,143]. These studies thus clearly suggest that artificial electrical light and electronic gadgets have contributed significantly in inducing sleep debt and a steady decline in sleep amount and as a result health, in humans.

In addition, chronic illnesses such as depression, obstructive sleep apnea, and metabolic disorders also influence sleep quality and contribute to sleep deprivation.

It is, therefore, essential to understand the architectural sleep patterns and attainment of their minimum daily amount in adults. Further, it is necessary to understand the clinical aspects of sleep deprivation and ways to prevent it. Here, we review the consequences of sleep loss on overall health and quality of life in young and old adults.

Sleep architectural patterns during one night of sleep across different ages

During the initial days, it was strongly believed that sleep was a passive mechanism and was generated because of a mechanical blockage in nerve conduction. Due to the wide acceptance of von Economo's findings in the 1950s that the lesion in the anterior hypothalamus of the brain causes insomnia and the posterior hypothalamus causes hypersomnia, the passive theory of sleep was completely disregarded. The studies from the early 1960s to late 1970s categorically demonstrated the sleep-wake circuitries in the brain. There are multiple wake centers located in the brainstem, midbrain, and basal forebrain; a solitary ‘ventro-lateral preoptic area’ (VLPO) located in the anterior hypothalamus involved in non-Rapid Eye Movement (NREM) sleep generation and Rapid Eye Movement (REM) sleep ‘ON’ and ‘OFF’ centers located in the brainstem [71]. The various brain areas interact with each other in a coordinated fashion to modulate sleep and wakefulness. Brain activity during different stages results in the formation of distinctive electroencephalogram (EEG) patterns. An EEG pattern specific to each stage of sleep and waking can be utilized to determine the stage of sleep.

Allan Rechtschaffen and Anthony Kales committee constituted by the ‘American Academy of Sleep Medicine (AASM)’ in 1967 developed a protocol, which is popularly known as the “R and K rules”, to characterize sleep stages in healthy adult subjects [117]. The “R and K rules” proposed four stages of NREM sleep Stage-1, 2, 3, and 4 and a single REM sleep stage as Stage 5. Further, the R and K rules were modified and updated in 2007, which is now known as the “AASM scoring manual” [17]. The AASM scoring manual recommended four sleep stages instead of five stages in humans. These stages are N1 (previously S1), N2 (previously S2), and N3 (the previous S3 and S4 stages were merged into N3), and stage R sleep (REM sleep) [17].

Electroencephalogram (EEG) in humans exhibits low amplitude and desynchronized wave pattern with increased activities of alpha and beta waves, high muscle activity in electromyogram (EMG), and rapid eye movement activity in electrooculogram (EOG) during wakefulness. The stage ‘N1-NREM sleep’ is a transitional state between wake and drowsiness. EEG during the N1 stage exhibits low amplitude and fast-wave activity, but 50 % of epochs show theta wave (4–7 Hz) activity. The EMG shows relatively less activity than the wakefulness, and the eyes begin to show slow rolling eye movements (SREMs). Stage ‘N2-NREM sleep’ is intermediate sleep and is characterized by a predominant appearance of EEG theta activity (4–7 Hz). During this sleep stage, spindles and K complexes start appearing in a typical episodic manner.

However, during N2, K complexes essentially remain present with or without sleep spindles. Stage 'N3-NREM sleep' or slow-wave sleep (SWS) or delta sleep is marked by high-amplitude slow delta waves and decreased muscle tone. Both K complexes and sleep spindles remain present in stage N3 sleep.

Stage 'R' or REM sleep is characterized by EEG epochs with low-amplitude, mixed-frequency waves along with low or no EMG tone, with active eye movements. Sometimes eye movements do not appear during this stage, but the amplitude of EEG remains low along with mixed-frequency waves. In the majority of 'R' epochs, the eyes move rapidly under closed eyelids and also during dreaming, but it remains absent in some of the episodes. Hence, eye movement essentially not be needed to mark the REM sleep periods.

Sleep in young adults

Majority of young population, particularly school and college-going young adults, face sleep deficits. A study conducted on the health behavior of school-aged children (15 years) in England reports that 30 % of boys and 49 % of girls exhibited sleeping disturbances. Among them, 36 % students admitted that they were not getting enough sleep and were unable to focus on their schoolwork [26]. Inadequate sleep during the adolescent age can lead to long-term sleep issues which can even persist during old age. A group of subjects had sleep disturbances during their adolescent age, which persisted in 1/3rd of subjects till their 23rd yr and in 1/10th of subjects till the age of 42 yrs [43]. Additionally, compared to 36 percent of teenagers and adults, the majority of college students exhibited daytime sleepiness [100,29]. The most frequent symptoms were tiredness upon waking (46 %), followed by anxiety and overall weakness in teenagers, excessive drowsiness (50 %) followed by nervousness and tension in college students, and tension (49 %) followed by nervousness and irritability in office goers or workers. The connections between sleep demand, sleep index, fatigue, and mood was more pronounced in younger respondents. Surprisingly, indications of tiredness were prevalent in college students, school-age children, as well as in office-goer adults. Many factors, including the dysfunctional circadian machinery, can be attributed to sleep disruption [100,29]. It appears that women are more severely impacted by chronic sleep loss than males. Forty-seven percent of women report having trouble sleeping, mostly due to sleeping for <6 h per night [133]. The prevalence of insomnia complaints in postmenopausal women was examined using polysomnography recording, which ranged from 61 to 83 percent [33]. The situation becomes more serious with clinical conditions such as pain, obesity, pregnancy, etc.

Sleep during pregnancy and postpartum

Several studies have suggested that sleep considerably gets altered during and after pregnancy. Many physiological, societal, and psychological modifications arise at some

stage in the perinatal and postpartum intervals, potentially modifying sleep. Amongst many, the postpartum infant-care duty and postpartum melancholy can be the elements that likely give the mother sleepless time for days. Animal models are best suited to study the effect on sleep loss during pregnancy and postpartum. Many similar studies have been conducted on pregnant mammals like whales and dolphins. In these species too, mothers as well as calves, both remained active for the entire day at some point of the first postpartum month and did not show slumbering behavior in any respect, primarily due to an external risk [86,129,85]. The aquatic mammals such as whales and dolphins exhibit unihemispheric sleep (one half of the brain sleeps while the other half remains active). Unihemispheric sleep can visually be observed as one eye (contralateral to the sleeping brain) remained closed while the other eye (ipsilateral to the sleeping brain) remained open. In preliminary postpartum months, the mothers did not display either uni- or bi-lateral eye closure at any stage. The calf, however, showed unilateral eye closure one month after birth [85]. It appears that the mother and calf had been almost sleepless for a month. However, it is not clear yet if such sleepless behavior is unique to an aquatic environment only or if other mammalian animals also exhibit such sleepless behavior.

Unlike the cetacean mother and calf, however, in the laboratory, mother rats/mice and their pups, mother cats, and their kittens did not show sleepless behavior at all. Although, during pregnancy and postpartum, the Wistar rats showed frequent intermittent arousal from sleep, but NREM sleep and delta power during NREM sleep significantly increased [134]. Nevertheless, in the third trimester of pregnancy, the anxiety level increased in the mother rats, which gradually improved after parturition [134]. It is worth mentioning that the mother rat modified and synchronized its sleep-wake timing with the activity of suckling. For example, the mother rats mostly remain in NREM sleep when pups suckle. However, the sleep of mothers remains fragmented during nursing. Interestingly, the delta sleep load did not change, rather it exhibited a load comparable to the non-nursing periods [16]. Overall it appears that pregnancy and postpartum periods alter the normal sleep-wake cycle, but at the same time, some compensatory re-appropriation also takes place.

In humans, pregnant and postpartum women also experience sleep deficits. For example, nighttime waking, intermittent arousal, frequent daytime napping, poor sleep quality, snoring, sleep apnea, etc., have been noticed in the majority of pregnant women [103,53]. Studies suggest that old adult mothers have a higher prevalence of poor sleep quality than young mothers [161]. Although the precise cause of sleep deficit in pregnant women is not known, evidence suggests that hormonal changes during pregnancy could be the primary factor [135]. In addition, significant bodily changes during pregnancy, for example, fetus demands for womb space, respiratory and micturition discomfort to the mother because of an increase in womb volume, postural discomfort, etc., could be other factors that contribute to causing sleep distress. These factors may contribute to developing depression, anxiety, and antepartum suicidal tendency in some women during pregnancy and

the postnatal period [60]. Also, late-sleeper pregnant women are highly sensitive to developing postpartum mania, depression, and obsessive-compulsive disorder [99]. Obeyesekere et al., in one of their studies, divided the participating subjects into “early and late sleeper” groups based on their sleeping lifestyle. The early sleepers go to bed around 11:30 PM, while the late sleepers sleep after 11:30 PM. They reported that the late sleeping women had a higher prevalence of developing symptoms of mania, depression, and obsessive-compulsive disorder during the postpartum period [99]. In another study, King et al. reported that poorly sleeping mothers exhibited significantly less maternal caregiving attitudes toward infants [76]. From these studies, it appears that pregnancy and postpartum conditions influence sleep quite adversely. Although the compensatory processes attempt to re-appropriate the sleep loss, if not compensated by any chance, it has long-lasting consequences on the child's health.

Sleep and sleep disturbances with normative aging

Sleep is vital for the maintenance of our normal health and cognition [87,34,88,146,78,112,148,113,147,149]. Unfortunately, sleep is altered in subjects with normative aging both qualitatively as well as quantitatively (Fig. 1). The energy of the sleep oscillatory waves, sleep efficiency, total sleep time, NREM and REM sleep ratio, NREM and REM sleep episode length are significantly altered with aging [91,80]. The noticeable changes in sleep architecture one can observe with aging are early bedtimes and rise times, sleep fragility, increase in N1 and N2 sleep stages and decrease in N3 sleep amount, etc. [150,101,118,38]. In addition, sleep urge and daytime sleepiness are also prominent in older subjects [57,154].

The aged subjects experience an alteration in circadian oscillation. For example, the circadian timing of body tem-

perature rhythm and melatonin secretion in elderly subjects are altered, and it could be the causal factor of sleep disturbances in old adults [42,96]. In addition, the wake-inducing circadian signal becomes weak in the aged person, especially during the evening, and sleepiness increases during the wake period (late afternoon) [140]. The old adults also exhibit a preference for morningness, a phenomenon that is associated with the circadian phase advancement [28]. Besides these, NREM sleep amount, along with the amplitude of delta waves and evoked K-complex, decrease in a linear fashion with normative aging. These studies suggest that there could be a gross alteration in homeostatic as well as circadian regulation of the sleep-wake cycle with aging [19,37].

Sleep also modulates the hormonal secretion of the neuroendocrine system. For example, the pituitary gland secretes and releases growth hormone only during sleep. Aging significantly influences sleep amount, which decreases exponentially from adulthood to middle age (approx 50 years) to late age (approx 80 years). It is, however, not known if the changes in sleep fragmentation and NREM sleep amount from younger to older adults are associated with a parallel decrease in growth hormone secretion [150]. But the release of human growth hormone during sleep is significantly related to the synchronized stages of sleep, suggesting that these two are closely linked through a common neural mechanism [125]. Further, Cauter et al. have observed that the elevated evening level of cortisol could be associated with an age-related decline in the REM sleep amount [150]. The release of another pituitary hormone, thyroid-stimulating hormone (TSH), is also associated with the sleep-wake cycle. The level of TSH remains low during the daytime but starts increasing in the late afternoon and attains a peak at sleep onset. The level of TSH declines gradually at night and returns to its low level during the day [83]. Since TSH secretory rhythm is tightly regulated by the sleep-wake pattern, hence, it is quite

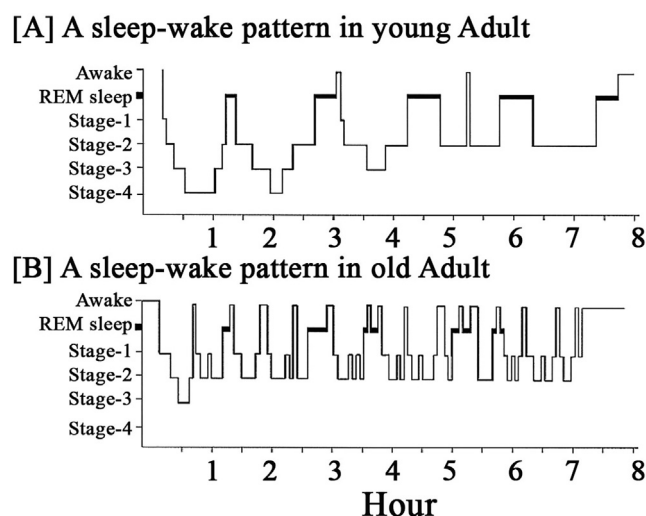


Fig. 1. The changes in sleep architecture in an old adult are shown in comparison with a young healthy adult. The hypnogram is taken and reproduced with permission and licensed from the paper entitled “*Insomnia in the Elderly: Cause, Approach, and Treatment*” by Licensed Content Authors Nabil S. Kamel, and Julie K. Gammack in the journal *The American Journal of Medicine*. (License Number: 5416321461088; License date: Oct 26, 2022).

obvious that an age-associated altered sleep-wake cycle could influence the level of circulating TSH, which may, in turn, cause subclinical hypothyroidism and homeostatic ionic imbalance in old adults [1,93]. Subclinical hypothyroidism causes cardiovascular and cognitive dysfunctions, depression, and disability [1]. It is highly possible that age-associated physiological, cognitive, and bodily disability could be associated with sleep-deficit-mediated altered circulatory TSH levels in older adults. These studies altogether suggest that aging-associated decrease in health and cognition could mainly, if not exclusively, be due to poorer sleep quality [1,58]. It is intriguing to ask that if a gradually decreasing sleep amount with aging is a normal phenomenon, then how and why does the quality of life gets compromised in older adults? We need to understand clearly that the sleep architecture changes as we grow, but frequent arousal, waking up tired almost every morning, and poor sleep are not part of normal aging.

Sleep is invariably getting disturbed due to the pressure of performing societal and professional duties in this modern world. Our brain and body undergo several physiological and behavioral changes because of sleep deprivation. Numerous health issues, including metabolic dysfunction, hypertension, cerebral vascular disorders, diabetes, neurocognitive diseases, and cardiometabolic consequences, as well as increased mortality, may be caused by insufficient sleep, be it either sleep quantity or quality, or both. Besides these, there are over 80 different sleep disorders that have been reported, but the most common are hypersomnia, circadian rhythm abnormalities, sleep-related movement disorders, obstructive sleep apnea, insomnia, etc. As adults get older, sleep problems begin, and maintenance of healthy sleep habits become more difficult. Increased sleep latency (the amount of time needed to fall asleep), sleep fragmentation (the frequency of nightly awakenings), the length of awakenings, and a decrease in the total amount of sleep are symptoms that represent these challenges (Fig. 1). Some of the common sleep-associated diseases, which are quite prevalent in our society across ages, are deliberated here (Table 1).

Sleep disorders in young and old adults

Insomnia, excessive sleepiness, and abnormal events that occur during sleep are the diagnostic features of sleep disorders or somniphobia (Table 1). Also, sleep is commonly altered in several diseases. For example, poor sleep quality and chronic fatigue are common among patients suffering from chronic diseases such as rheumatoid arthritis, fibromyalgia, etc. Besides, sleep deficiency itself is the root cause of several diseases. Hence, it is essential to have a complete understanding about sleep disorders and sleep alteration in various diseases.

Insomnia, parasomnia, and hypersomnia

Excessive difficulty in falling asleep is termed 'insomnia.' It includes difficulties in sleep initiation or its maintenance, prolonged wakefulness, sleep fragmentation, etc. Insomnia is characterized as primary and secondary

insomnia. Primary insomnia could be attributed to sleep hygiene or behavioral or adjustment factors, whereas secondary insomnia is primarily due to medical or psychiatric illness or substance abuse [49,144]. Parasomnias include sleep disturbances primarily due to some physical or emotional sporadic events, such as emotional and behavioral incitement, intense dreaming, etc. These events are not frequent, but rather occur occasionally. Few behavioral parasomnias are violent and associated with dream enactment. The negative dream enactment associated with violent fighting potentially causes injury to the sleeper or bed partner. Such episodes may initially occur about once a week but may go up to four times a night [56].

Hypersomnia is characterized as excessive daytime sleepiness and prolonged nocturnal sleep episodes, with more than 9 hrs of daily sleep [119]. Hypersomnia can be characterized as idiopathic and behavioral hypersomnia. In idiopathic hypersomnia, patients exhibit a remarkable increase in sleepiness, with prolonged nocturnal sleep without REM sleep alteration [7]. Behavioral hypersomnia includes 'Behaviorally Induced Insufficient Sleep Syndrome (BISS),' where subjects voluntarily restrict night-time sleep for their job, study, or some personal activities such as watching TV, etc. BISS is one of the leading causes of health issues such as fatigue, mood disorder, excessive eating, and weight gain, especially in middle-aged people [158].

Kleine-Levin syndrome and menstrual-related hypersomnia are also categorized as hypersomnia. In Kleine-Levin syndrome, subjects exhibit symptoms such as; intermittent hyper-somnolence, hyperphagia, cognitive deficits, hyperactivity, and hypersexuality [115]. Menstrual hypersomnia is also marked by hypersomnolence a few days before the onset of the menstruation cycle. The actual cause of the disease is not known, but the hormonal alteration during the menstrual period could be the cause of excessive sleepiness [114].

Restless leg syndrome (RLS) and periodic limb movement disorder (PLMD)

Patients suffering from restless leg syndrome (RLS) feel an intense urge to move their leg during sleep, mainly due to unpleasant sensations or pain in the affected leg [144]. The periodicities of limb movement increase by evening and get severe during sleep. Although RLS and periodic limb movement disorder (PLMD) appear very similar, they are in fact quite different [65,144]. The PLMD patients move one and/or both legs rhythmically several times a night, but they remain unaware of the condition. In comparison, the RLS patients exhibit a strong urge for the movement of only one leg as they either feel pain or an uneasy sensation, tactile hallucination, or a feeling of formication before or at sleep onset [144]. Because of pain and recurring limb movement, sleep gets highly disturbed, and ultimately the patients seek medical intervention. Increasingly, the incidences become worse with aging. There is 9–20 % and 4–11 % prevalence of RLS and PLMD in old adults, respectively [65,144].

Table 1

Sleep disorders, their general symptoms, and changes in sleep architecture.

Sleep Disorders		Sleep Architectural Changes	General Symptoms
1. Insomnia	(a) Primary InsomniaSome intrinsic/extrinsic factors or both are associated with its etiology (b) Secondary insomniaAs consequences of a medical or psychiatric illness, another sleep disorder, or substance abuse	Both NREM and REM sleep	• Repeated sleep-initiation difficulties, Decreased sleep duration • Reduced sleep consolidation, or quality even though the patients get adequate time and opportunity to sleep • Insomniac patients may tend to wake-up early and no signs of restorative sleep
2. Hypersomnia	(a) Kleine-Levin syndrome (b) Menstrual-related hypersomnia (c) Idiopathic hypersomnia (d) Behaviorally Induced Insufficient Sleep Syndrome (BISS) or Insufficient sleep syndrome	Both NREM and REM sleep	• Primarily excessive daytime sleepiness • Nocturnal sleep remains normal • Behaviorally chosen nocturnal sleep restriction to perform activities such as job, study, watching TV, etc
3. Parasomnia	1. Disorders of arousal from NREM Sleep (a) Confusional arousals (b) Sleepwalking (c) Sleep terrors 2. Parasomnias associated with REM sleep (a) Rapid eye movement sleep behavior disorder (RBD) (b) Recurrent isolated sleep paralysis (c) Nightmare disorder Other Parasomnias (a) Sleep-related dissociative disorders (b) Sleep enuresis (c) Sleep-related groaning (d) Exploding head syndrome (e) Sleep-related hallucinations (f) Sleep-related eating disorder	Both NREM and REM sleep Mostly REM sleep Mostly REM sleep	• Sleep state accompanied by some physical or emotional sporadic events, such as abnormal sleep-related movements, emotional and behavioral incitement, intense dreaming, etc. • These disorders may emerge from either NREM or REM sleep • Some of the parasomnias may appear due to drug or substance abuse or medical conditions • In sleep-related eating disorders (SRED), patients exhibit recurring episodes of eating during the transition from sleep to arousal
4. Narcolepsy	A. Narcolepsy with cataplexy B. Narcolepsy without cataplexy	REM sleep	• Associated with excessive daytime sleepiness • Hypersomnia is the most common symptom of narcolepsy • Sleep attacks occur despite having sufficient nocturnal sleep • Daytime sleep occurs daily, recurring typically every 2 hours (can vary) • Cataplexy is defined as sudden and transient episodes (< 2 minutes) of loss of muscle tone • It is generally bilateral and triggered by emotions (usually laughing and joking)
5. Circadian Rhythm Disorders	A. Advanced sleep-phase syndrome B. Delayed sleep-phase syndrome	Both NREM and REM sleep	• The sleep-wake cycle is misaligned with circadian timing • The patients complain of either insomnia or excessive sleepiness
6. Sleep Apnea	A. Obstructive Sleep Apnea B. Central Sleep Apnea	Both NREM and REM sleep	• The upper airway pathway is obstructed, which causes increased effort in breathing and inadequate ventilation during sleep • In central sleep apnea, respiration is altogether stopped intermittently or cyclically during sleep because of dysfunction in neural circuitries of the central nervous system
7. Restless Leg Syndrome / Periodic Limb Movement Disorder		Both NREM and REM sleep	• A very painful sensorimotor neurological sleep-associated disorder • The subjects feel an extreme urge to move extremities, mostly the legs, during sleep as the subjects feel unpleasant sensations or pain in the affected limbs • The periodicities of leg movement increase by evening and worsen during sleep

Sexsomnia (Sleep sex)

Sexsomnia is diagnosed with atypical sexual behavior during sleep, which falls under the category of parasomnia [47,15]. Sexsomnia patients usually have a family history of parasomnia and sleep-walking; hence, it is also linked to sleep-walking [47,15]. Sexsomnia is a very unusual kind

of parasomnia and is most likely associated with NREM sleep [47,44]. Sexual behavior during sleep varies, varying from simple sexual dirty talk (sleep talking) going to the extent of sexual assault. Interestingly, in one of the court proceedings, a three times rape-offender was acquitted by the court in England after considering his medical condition that he was suffering from 'sexsomnia' [94].

Sleep Apnea

Sleep apnea patients demonstrate breathing interruption while asleep. There are two categories of sleep apnea (a) Obstructive sleep apnea and (b) Central sleep apnea [141]. The obstruction in the upper airway pathway during sleep causes obstructive sleep apnea, which increases breathing effort and inadequate ventilation. The breathing obstruction during sleep causes a reduction in blood oxygen level, and the subjects experience frequent arousal [141,5]. The neuronal circuitries dysfunction in the central nervous system induces central sleep apnea, which is further characterized into two forms: (a) Primary central sleep apnea and (b) Secondary central sleep apnea [48]. The primary central sleep apnea is mainly characterized by the recurrent cessation of breathing episodes during sleep, and subjects hardly put effort into breathing (apneic episodes ≥ 5 per hour of sleep; $\text{PCO}_2 \geq 45$ mm Hg) [5,48]. Long-term consumption of opioids or other substances of abuse suppresses respiration by acting on the medullary respiratory neurons and causes secondary central sleep apnea. Some premature infants exhibit sudden respiratory pauses for 20 sec or more, and such sleep apnea is usually called apnea of prematurity. Further, it has been noticed that few infants also suffer from almost similar sleep apnea, and the reasons are primarily not known. Such apnea is called apnea of infancy [46,5].

Narcolepsy

The most common symptom of narcolepsy is also excessive daytime sleepiness and hypersomnia. The excessive urge to sleep occurs despite having sufficient nocturnal sleep. The periodic daytime sleep recurs approximately every 2 h. Sleep recurrences are often irresistible. Even having so much sleep load during the daytime, sleep duration is usually short and frequently associated with dreaming. Interestingly, a short nap offers a considerable refreshing value, except in children, who are often tired of waking, and severe sleepiness can also lead to unconscious micro-sleep episodes [14]. Narcolepsy can often be associated with a sudden and transient loss of muscle tone (episodes may last approx 2 min), and it is termed narcolepsy with cataplexy [18,14]. The patients having narcolepsy with cataplexy exhibit excessive daytime sleepiness with occasional episodes of “status cataplecticus,” which may last several minutes to hours. The loss of lateral hypothalamic hypocretin neurons is believed to be the main cause of narcolepsy (<110 ng/L of hypocretin-1 concentration in the CSF). It has also been found that the hypocretin levels remained normal in $< 10\%$ of narcolepsy with cataplexy patients. Therefore, some other factors may also be associated with the cause of narcolepsy.

REM sleep behavior disorder (RBD)

Rapid eye movement sleep behavior disorder (RBD) is a parasomnia characterized by dream enactment activity and loss of muscle atonia during REM sleep [127,130]. The person experiencing these symptoms and their bed companions could suffer catastrophic consequences. The majority of patients describe their dreams as nightmares,

and the content of these dreams frequently involves being chased or attacked by animals, insects, or humans [130]. The motor activity in RBD patients frequently starts with some repetitive jerking and is later followed by a more dramatic and seemingly intentional activity, such as punching, flailing as if to defend oneself, running, jumping out of bed, etc [130]. Patients and their bedmates get injuries as a result of these practices. RBD could be triggered by abnormal connections between the cortex and the brainstem, which regulate muscle tonicity, and therefore it is also referred to as REM sleep without muscle atonia [22,106]. Although, the definition of REM sleep without atonia and its clinical implications in people who have never engaged in dream enactment behavior are up for debate; nevertheless, these patients demonstrate “preclinical” or “subclinical” symptoms of RBD [22,106].

The cause of RBD is not fully known but elderly age, male sex, narcolepsy, antidepressant use, and neurological illnesses are risk factors that exacerbate RBD [21]. Neurodegenerative synucleinopathies, such as Parkinson's disease, dementia with Lewy bodies, olivopontocerebellar degeneration, multiple-system atrophy, Shy-Drager syndrome, narcolepsy, etc could be involved in the development of idiopathic RBD [21]. RBD caused by drugs is quite common in people using antidepressants. Serotonin reuptake inhibitors, tricyclic antidepressants, and monoamine oxidase inhibitors are the antidepressants that are most likely to cause an RBD episode. Studies have also indicated links of RBD to congenital and neurodevelopmental issues, traumatic brain injury (TBI), post-traumatic stress disorder (PTSD), and congenital disorders [21].

Phase advance or phase delay sleep disorders

Usually, the sleep-wake cycle is completely aligned with circadian timing, but when it is untied, patients are unable to sleep at actual bedtime. The patients either experience insomnia or excessive sleepiness. If the sleep periods are always delayed in comparison to normal sleep timing, such a condition is called ‘the phase delay sleep disorder.’ It is very common in adolescents or young adults [156,126]. If the sleep periods always occur before the actual bedtime, such a condition is called ‘the phase advance sleep disorder’, and it is very common in older adults [95,123]. The circadian cycle-associated sleep disorders are caused possibly because of insufficient daylight exposure, which plays a significant role in re-setting the biological clock [144]. The rapid changes in the time zones cause ‘Jet lag disorder’, which also influences sleep. The temporal incompatibility between sleep-wake timing and the circadian clock primarily induces sleep disruption [8,122]. Similarly, working in the night shifts also alters sleep-wake timing, and the subjects usually complain of insomnia or excessive daytime sleep [2,160].

Neuropsychological disorders and sleep alteration in young and old adults

Sleep architecture is altered in several psychological, physiological, and metabolic disorders. The magnitude of

sleep disturbance and relapse of some of the diseases are significantly positively correlated. For example, severe insomnia is a long-lasting symptom of depressive patients, and patients remain at a higher risk of subsequent depression [98]. Sleep management helps improve the anxiety and mood of patients with major depressive disorder (MDD) and generalized anxiety disorder (GAD) [110,137]. Sleep management has also been recommended as therapeutic to reduce the risks of frailty in the elderly [107,108]. Therefore, it is essential to understand the sleep disturbances in patients suffering from some chronic diseases such as anxiety disorders, metabolic disorders, neurological disorders, etc.

Anxiety, negative thinking, and sleep

Sleep management through clinical interventions improves anxiety and mood in depressive patients. Hypnotic drugs, along with antidepressant treatment, help improve depressive symptoms in severely depressive patients more than antidepressant drugs only [120]. Hence, medication/treatment for improving sleep quality could be a promising tool for curing anxiety and depression [54]. The pre-treatment of insomnia symptoms in some depressive patients is believed to be linked with poorer treatment outcomes of a single antidepressant drug [142]. Further, Dr. Michael Irwin's group has proposed that the risk for depression could be more in the elderly population if they have a prior history of the disorder. However, for those having a prior history of depression, post-treatment sleep disturbance could predict a relapse/recurrence of depressive disorder [82]. Hence, sleep optimization seems to be the best tool for the treatment of depression, which could further prevent the progression of depression in patients when they grow old [82]. Sleep disturbance is also a risk factor for the development of depression in HIV-infected patients. The chronically sleep-disturbed HIV-infected patients are more depressed than the HIV-infected patients having normal sleep amounts [67]. Targeted treatment of insomnia thus may possibly decrease depression in HIV-infected patients.

Sleep deprivation is associated with the generation of a negative emotional experience [75]. It has been observed that sleep deprivation alters the functional connectivity within the reward network and cerebral cortex. Total sleep deprivation decreases functional connectivity between the left nucleus accumbens (NAc) and anterior cingulate cortex (ACC) and between the right NAc with the ACC and right inferior frontal gyrus (IFG) [163]. Surprisingly, decreased functional connectivity negatively correlated with the scores of emotional experience [163]. Since sleep deprivation significantly altered functional connectivity in the network of the 'feel-good' events, that is why the sleep-deprived subjects may experience enhanced negative emotional thoughts after sleep deprivation [163].

Reactive aggression and sleep

Sleep loss issues have been investigated in clinical studies as a possible cause of reactive aggression and violence [124]. Several studies in animals and humans have found

a link between insufficient sleep and reactive hostility (Hsu et al., 2009, Jha and Mallick, 2009, [73,89]. Sleep deprivation does not cause physical violence in all people; nonetheless, psychiatric patients are particularly vulnerable to emotional outbursts when they are deprived of sleep. Sleep deprivation may alter the functioning of the prefrontal cortical neural network, which may, in turn, induce aggressive behavior in sleep-deprived subjects [73]. In addition, sleep disorders, anger, and violence in psychiatric patients may be connected to changes in the central serotonergic and hypothalamic–pituitary–adrenal axis [69,12]. As a result, it has been suggested that identifying at-risk persons and providing sleep management therapy may minimize aggressive and violent occurrences in the sleep-deprived subjects [73].

Frailty status in old adults and sleep

Frailty is the primary cause of increased hospitalization, falls, disability, and mortality rates among the elderly [32,52]. A major bodily system deteriorates, if not all of them, with frailty, including the musculoskeletal, cardiovascular, and neurological systems, to name a few. Hence, it is considered to be a crucial component in the development of disability in aged people [40].

The circadian rhythm, as well as the sleep cycle, is disrupted in the elderly. Daytime light exposure and sleep amount are both reduced in older persons [128]. Reduced sun exposure affects nocturnal pineal melatonin levels, which affects both circadian function and the sleep-wake cycle [151]. Decreased sleep quality and sleep efficiency, excessive daytime sleepiness, and sleep apnea are all factors that contribute to a high prevalence of frailty syndrome in the elderly [51,152,50]. It has also been observed that the elderly's frailty level is linked to longer sleep latency and daytime sleepiness [4]. It is, therefore, likely that the circadian system instability in the elderly may affect the sleep-wake cycle, which ultimately may induce frailty.

Furthermore, as people get older, their biological metabolic pathways change. Hormonal imbalance, a weakened immune system, oxidative stress, and psychological disorders are the first visible signs of aging. Growth hormone and testosterone outputs decline as people age, the pro-inflammatory system becomes more active, and the rate of oxidative stress rises. In addition, both chronic sleep deprivation and aging have an impact on the hypothalamic–pituitary–adrenal and hypothalamic–pituitary–gonadal axes activities [23,61,92,59,81]. Cortisol, testosterone, and estrogen levels are reduced by a disrupted sleep-wake cycle, whereas progesterone, prolactin, corticosterone, and ACTH levels are increased [6]. Therefore, it appears that the increased risk of morbidity and death associated with frailty could be closely linked to sleep disruptions in elderly people [13,104,55].

A sedentary lifestyle contributes greatly to frailty and sleep disruption [41,97]. To develop treatments to avoid frailty is a clinical priority. Physical activity, physical activity combined with nutrition, physical activity combined with nutrition and memory training, home modifications, pre-habilitation (physical therapy including exercise and

home modifications), and comprehensive geriatric assessment (CGA) have all been shown to reduce frailty and its consequences in the elderly, [111]. Clinical therapies for sleep disorders, such as CPAP therapy for OSA, bright light therapy, and other insomnia techniques, have also been advocated as medicines to minimize the risks of frailty in the elderly [107,108].

Alzheimer's disease and sleep

People with Alzheimer's disease (AD) and other dementias frequently experience changes in their sleep patterns [164,74,121]. They may wake up frequently during the night and have difficulty returning to sleep. These sleep issues are assumed to be the outcome of disease-related brain alterations that influence the sleep-wake cycle [164,74,121]. According to the recent findings, adults in their 50's and 60's who receive less than six hours of sleep are more likely to get dementia later in life. People who got less sleep each night were 30 percent more likely to be diagnosed with dementia than those who got regular sleep (defined as 7 h) [121].

Alzheimer's disease patients frequently have sleep disturbances at night. Clinically, this can manifest as distress during the night, which can impact upto a quarter of AD patients at some point during their illness. Sleep disturbances in AD patients are complex, involving sleep-disordered breathing and altered chronobiology, both of which lead to excessive daytime napping. Polysomnographically, AD patients have less REM sleep in proportion to the severity of their dementia [20]. It has been observed in the 3xTg-AD mouse model of Alzheimer's disease that persistent disruption of the daily sleep-wake pattern promotes brain amyloid-beta ($A\beta$) levels and neuro-inflammation. Female 3xTg-AD mice were subjected to chronic sleep fragmentation, which consisted of four daily sessions of enforced wakefulness (one hour each) uniformly spaced during the light phase, five days a week for four weeks. Sleep fragmentation changed the daily sleep-wake rhythm to mimic the pattern seen in Alzheimer's disease. The levels of amyloid-beta, **A β 40 and A β 42**, in hippocampus tissue from sleep-fragmented animals were greater than in undisturbed controls. The levels of tau and phospho-tau in the hippocampi of sleep fragmented and undisturbed control mice, on the other hand, did not change much [45].

Memory deficits and sleep disturbances are hallmarks of both aging and Alzheimer's disease. Rauchs et al. have looked at the possible connection between these abnormalities, focusing on sleep spindles, which are involved in memory consolidation [116]. They offered two episodic memory tasks to young and old healthy volunteers, as well as AD patients, and recorded sleep after learning. They observed that sleep spindles decreased universally in aging and AD patients. Fast spindles were reduced in AD patients only. Interestingly, the mean intensity of fast spindles was strongly connected with the instant recall performance of the AD patients. The study showed that a specific decrease in rapid spindles in AD patients is linked to their learning capacities [116]. All these studies suggest that sleep alter-

ation in AD patients might be associated with the manifestation of disease severity.

Parkinson's disease and sleep

Sleep disturbances are the second most common complaint among people with Parkinson's disease, affecting 64 percent of patients (ranging from 41.1 to 78.3 %) [84]. Sleep disturbances in PD could be multifactorial and might be present both during the day and at night. The presence of concurrent sleep disturbances such as REM sleep behavior disorder, restless legs syndrome, or breathing disorders such as obstructive sleep apnea is linked to the motor (akinesia, rigidity, and dystonia) and autonomic systems (nocturia) dysfunction [9]. Excessive daytime sleepiness and sudden sleep attack are diurnal manifestations that can result from nocturnal sleep impairment, dopaminergic therapy, or, more intriguingly, from the neurodegenerative process of Parkinson's disease itself [84].

Patients with Parkinson's disease exhibit a wide range of sleep disorders [145,136]. For example, dysregulation of sleep and wakefulness lead to insomnia and daytime drowsiness. Alteration in control of motor activity during sleep results in the development of parasomnias, mainly REM sleep behavior disorders, sleep-walking, overlap parasomnia, etc. Restless leg syndrome has also been reported to be common in Parkinson's disease patients [145,136]. In PD patients, the circadian machinery is somehow altered. It has been observed that there was no time-dependent variation in the expression level of the *Bmal1* clock gene in the suprachiasmatic nucleus (SCN) [25]. Also, reduced SCN neuronal activity with the overexpression of α -synuclein has been noticed in PD patients [77]. It suggests the weakening of master clock circuitries in the PD patients, which can ultimately influence many functions, including the S-W cycle. The circulating hormone profiles, such as the decreased melatonin production and the hypothalamic dysfunctions, are also noticed in PD [79,24,109]. These might be the reasons that abnormal sleep macro-architecture such as increased sleep latency, decreased sleep efficiency, and REM sleep amount are observed in the PD patients.

Conclusion

In conclusion, sleep disturbance in children, young and old adults is a worldwide public health issue that is linked to a number of clinical diseases. A person may be sleep-deprived, whether on purpose or accidentally, if they don't get the suggested minimal daily amount of sleep. Teenagers and young adults, most likely and unknowingly, purposefully deprive themselves of sleep. Few subjects may accidentally miss out on getting enough sleep due to their duties and vocation. Other causes of sleep deprivation include depression, obstructive sleep apnea, hormonal abnormalities, and other long-term conditions. While there is no substitute for restorative sleep, various actions could mitigate its negative effects. Therefore, understanding the clinical implications and preventing chronic sleep debt requires a significant degree of care. It is a medical disease

that cannot be compromised, according to prevalence and morbidity data. All facets of society, including caregivers, health technologists, and the community as a whole, must pay attention to the importance of sleep and the detrimental effects of its loss.

Declaration of Competing Interest

The author declares that she has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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